Metadata: DIF-Traits_data.csv

General description

This dataset contains pot-level measurements from a three-year factorial pot experiment manipulating water amount, watering frequency, and plant composition to assess functional traits and fitness components of *Bromus erectus*. The dataset also contains selected counts, traits, biomass measurements, and fitness variables for *Inula salicina* in heterospecific pots.

Each row represents one retained experimental pot. Functional trait values are aggregated at pot level from multiple individuals; fitness components are reported at pot level or per seed as specified below. Detailed sampling and calculation protocols should remain linked to Tables S3 and S4 of the manuscript and Supporting Information.

The dataset contains 66 retained experimental pots. Fourteen Dry-High pots were excluded due to complete plant mortality.

Experimental factors

Variable	Description	Type	Observed values
SampleID	Unique sample identifier	Character	66 unique values
Pot	Pot number (experimental unit)	Character	66 unique values
Amount	Water amount treatment	Factor	Wet; Dry
Frequency	Watering frequency treatment	Factor	Low; High
Composition	Plant composition treatment: B. erectus with conspecific neighbours or with I. salicina	Factor	Conspecific; Heterospecific

Amount	Frequency	Composition	Rows
Wet	Low	Conspecific	10
Wet	Low	Heterospecific	10
Wet	High	Conspecific	10
Wet	High	Heterospecific	10
Dry	Low	Conspecific	10
Dry	Low	Heterospecific	10
Dry	High	Conspecific	3
Dry	High	Heterospecific	3

Variable dictionary

Species scope	All biological response variables without the suffix **_Inula** refer to *Bromus erectus*, except **TB_pot**, which represents total community biomass. Variables ending in **_Inula** refer to *Inula salicina* and are included only for heterospecific pots.

Growth traits

Variable	Description	Units
N	Ramet number per pot (2024)	count
H	Mean plant height	cm
SB	Shoot biomass per individual (per ramet)	g
RB	Root biomass per individual (per ramet)	g
TB	Total biomass per individual (per ramet)	g
R_S	Root-to-shoot biomass ratio	unitless

Leaf traits

Variable	Description	Units
SLA	Specific leaf area	$cm^2\ g^{-1}$
LDMC	Leaf dry matter content	$mg\ g^{-1}$
LCC	Leaf carbon content	%
LNC	Leaf nitrogen content	%
Leaf_CN	Leaf carbon-to-nitrogen ratio	unitless

Physiological traits

Variable	Description	Units
Chl	Chlorophyll content per leaf area	$mg\ m^{-2}$
PSIIeff	Efficiency of photosystem II photochemistry (Fv/Fm)	unitless
Ψosm	Leaf osmotic potential	MPa
SS	Stomatal size	µm
SD	Stomatal density	$No.\ mm^{-2}$

Root traits

Variable	Description	Units
RD	Mean root diameter	mm
RDMC	Root dry matter content	$mg\ g^{-1}$
SRL	Specific root length	$m\ g^{-1}$
SRA	Specific root area	$cm^2\ g^{-1}$
RTD	Root tissue density	$g\ cm^{-3}$
RCC	Root carbon content	%
RNC	Root nitrogen content	%
Root_CN	Root carbon-to-nitrogen ratio	unitless

Variable dictionary

Fitness components

Variable	Description	Units
SER	Seedling establishment rate (2022)	proportion (0-1)
SSR1	Seedling survival rate (2022-2023)	proportion (0-1)
SSR2	Seedling survival rate (2023-2024)	proportion (0-1)
SP1	Seed production per sown seed (2023)	count per seed
SP2	Seed production per sown seed (2024)	count per seed
SPtotal	Total seed production per sown seed (2023-2024)	count per seed
RP1	Ramet production per sown seed (2023)	count per seed
RP2	Ramet production per sown seed (2024)	count per seed
TBs	Total biomass per sown seed (2024)	g per seed

Biomass and plant count variables

Variable	Description	Units
TB_pot	Total community biomass per pot, including both species where present (covariate)	g
N2022	B. erectus seedling number per pot (2022)	count

Inula salicina traits

Variable	Description	Units
N2022_Inula	Seedling number per pot (2022)	count
N_Inula	Plant number per pot (2024)	count
Size_Inula	Mean plant diameter	cm
SB_Inula	Mean shoot biomass per plant	g
RB_Inula	Mean root biomass per plant	g
TB_Inula	Mean total biomass per plant	g
TB_pot_Inula	Total biomass per pot	g
R_S_Inula	Root-to-shoot biomass ratio	unitless
SLA_Inula	Specific leaf area	$cm^2\ g^{-1}$
LDMC_Inula	Leaf dry matter content	$mg\ g^{-1}$
Chl_Inula	Chlorophyll content per leaf area	$mg\ m^{-2}$
PSIIeff_Inula	Efficiency of photosystem II photochemistry (Fv/Fm)	unitless
SER_Inula	Seedling establishment rate (2022)	proportion (0-1)
SSR2_Inula	Survival rate (2023-2024)	proportion (0-1)